

Assignment-4

No. of machines = 3

machines = A, B, C

no. of plays = $N_{\text{play}} = 10$

Initial values =

$$N(A) = N(B) = N(C) = 0$$

$$R(A) = R(B) = R(C) = 0$$

$$Q(A) = Q(B) = Q(C) = 0$$

UCB:-

$$UCB(a) = Q(a) + \sqrt{\frac{2 \ln(t)}{N(a)}}$$

if $N(a) = 0$

$UCB(a) = \infty$ (This ensures that every machine is tried at least once)

② Incremental Sample-Avg formula:-

$$Q_{\text{new}}(a) = Q_{\text{old}}(a) + \frac{R - Q_{\text{old}}(a)}{N(a)}$$

Initial table

machine	N	R	Q	UCB
A	0	0	0	∞
B	0	0	0	∞
C	0	0	0	∞

Since all machines have $N=0$, each must be explored.

for play-1

$$R=3$$

$$N(A)=0$$

$$N(A)=0+1=1$$

$$R(A)=3$$

$$Q(A) = 0 + \frac{3-0}{1}$$

$$\therefore \boxed{Q(A)=3}$$

After play 1:-

machine	N	R	Q
A	1	3	3
B	0	0	0
C	0	0	0

Reward $R=6$

$$N(B)=0$$

$$N(B)=1$$

$$R(B)=6$$

$$Q(B) = 0 + \frac{6-0}{1} = \boxed{6 = Q(B)}$$

After play 2:-

A	1	3	3
B	1	6	6
C	0	0	0

$$R=4$$

$$N(C)=1$$

$$R(C)=4$$

$$Q(C) = 0 + \frac{4-0}{1}$$

$$\boxed{Q(C)=4}$$

UCB calculation

$$Q(A) = 3, N(A) = 1$$

$$Q(B) = 6, N(B) = 1$$

$$Q(C) = 4, N(C) = 1$$

for machine A:-

$$UCB(A) = 3 + \sqrt{\frac{2 \ln(4)}{1}}$$

$$\ln(4) \approx 1.3863$$

$$= 3 + \sqrt{2(1.3863)}$$

$$= 3 + 1.665$$

$$\boxed{UCB(A) \approx 4.665}$$

machine B

$$UCB(B) = 6 + \sqrt{\frac{2 \ln(4)}{1}}$$

$$= 6 + 1.665$$

$$\boxed{UCB(B) \approx 7.665}$$

for machine C

$$UCB(C) = 4 + \sqrt{\frac{2 \ln(4)}{1}}$$

$$= 4 + 1.665$$

$$\boxed{UCB(C) \approx 5.665}$$

$$\therefore UCB(B) > UCB(C) > UCB(A)$$

Hence, machine B is selected.

Incremental Q calculation:-

$$Q(B) = 6$$

$$R = 7$$

$$N(B) = 2$$

$$Q_{\text{New}}(B) = Q_{\text{old}}(B) + \frac{R - Q_{\text{old}}(B)}{N(B)}$$

$$Q_{\text{New}}(B) = 6 + \frac{7-6}{2} = 6 + \frac{1}{2} \\ = 6.5$$

A : 3, 4, 2, 5

B : 6, 7, 5, 8

C : 4, 3, 5, 4

machine A:- $Q(A) = \frac{3+4+2+5}{4} = 14/4 = 3.5$

machine B:- $Q(B) = \frac{6+7+5+8}{4} = \frac{26}{4} = 6.5$

machine C:- $Q(C) = \frac{4+3+5+4}{4} = \frac{16}{4} = 4$

$$Q(B) > Q(C) > Q(A)$$

Total rewards = $14 + 26 + 16$
 $= 56$